

Abstract

Web Scraping and Product Price Analysis

Product prices, discounts, and ratings on e-commerce platforms are scattered across many product pages, making it difficult to compare air conditioners, brands, and specifications through manual analysis. This project automatically collects Amazon India Air Conditioner listings through web scraping using Requests and BeautifulSoup, yielding 32 listings from 16 brands with details such as name, ratings, number of ratings, selling price, and actual price (MRP). Using Python, Pandas, NumPy, Matplotlib, and SciPy, the data is processed in an eight-stage pipeline: loading and reading, acquisition and filtering, extraction, validation and cleaning, aggregation, analysis, visualization, and interpretation. Tonnage, energy-star rating, brand, AC type, and inverter and Wi-Fi features are extracted from product titles; missing values are kept blank rather than filled with the median, six invalid MRP values (selling price higher than MRP) are removed, and three price outliers are flagged. Exploratory Data Analysis (EDA) using descriptive statistics, grouped comparisons, correlation analysis, and nine visualizations studies price distribution, brand-wise pricing, tonnage-wise trends, energy-star shares, discount patterns, and the relationship between price and ratings. The final analysis covers 32 listings, of which 31 have verified selling prices and 25 have verified discounts. The median price is Rs 34,990 and the average discount is 40.3%; price rises with tonnage ($r = 0.74$), and 5-star models cost about Rs 10,700 more than 3-star models. The project aims to support buyers and sellers in understanding AC price differences, discounts, brand trends, and product characteristics, while demonstrating an effective Python-based workflow for web-scraped e-commerce data analysis.

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